Gradient Flow Dynamics of Teacher-Student Distillation with the Squared Loss

Berkan Ottlik[1,2]

Flatiron Institute[1], Columbia University[2]

We study a teacher-student learning setup, where a "student" one layer neural network tries to approximate a fixed "teacher" one layer neural network. We analyze the population gradient flow dynamics in the previously unstudied setting with exactly and under-parameterization, even Hermite polynomial activation functions, and squared loss. In the toy model with 2 teacher neurons and 2 student neurons, we fully characterize all critical points. We identify "tight-balance" critical points which are frequently encountered in simulation and greatly slow down training. We prove that with favorable initialization, we avoid tight-balance critical points and converge to the global optimum. We extend tight-balance critical points and favorable initializations to the multi-neuron exact and under-parameterized regimes. Additionally, we compare dynamics under the squared loss to the simpler correlation loss and describe the loss landscape in the multi-neuron exact and under-parameterized regimes. Finally, we discuss potential implications our work could have for training neural networks with even activation functions.

BNOTE: Add a disclaimer that berfin and denny didn't read this yet.

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1 Introduction

We have two main motivations for pursing this work: (1) We are fascinated by neural network (NN) dynamics. This fascination stems from an intrinsic curiosity about complex systems and similarities to physical models such as the Ising model [cite]. NNs are particularly well-suited for study since, unlike many complex systems, we have full information about individual neurons and how they update. (2) We are optimistic that mathematically analyzing gradient-based training dynamics of NNs in key understudied regimes can make progress towards key NN research themes and ultimately make NNs more controllable and efficient [cite https://mishabelkin.substack.com/p/the-necessity-of-machine-learning]. Our work is most directly related to the research themes of loss and activation function choices, optimization, distillation, and superposition of features.

BNOTE: Qualify this by saying smth like some people make this distinction or a useful distinction is blah. Natural science research (and some deep learning theory) is typically theory driven. Researchers run experiments to find phenomena existing theories cannot explain and use experimental results to develop new theories. Machine learning research is typically methods driven. Researchers have hypotheses about what could provably or empirically improve model performance, that they test by proving theorems or testing on a validation dataset [cite mathematics of generalization, structure of scientific revolutions, 1988 NeurIPS things]. Our research is phenomena driven. We aim to understand the phenomena of a simple model and generalize the phenomena as much as possible. We hope our results are useful for theory and methods driven research in NNs.

We specifically focus on the under-parameterized teacher-student learning setup, where a "student" one layer NN tries to approximate a fixed "teacher" one layer NN. This setting allows us to directly study distillation and feature superposition. Additionally, it allows us to study learning any function that can be parameterized with a neural network.

1.1 Related works

Many works have studied gradient flow trajectories in the teacher-student setup. [Saad and solla] derive the gradient flow differential equations, yet rely on numerical integration to understand dynamics. [Du] gives convergence rates in the over-parameterized regime with the squared loss but only considers one student neuron. [Martin] considers over exact and under-parameterized regimes with the squared loss and multiple student and teacher neurons, but [martin] only considers the quadratic actiation function. [Berfin old and new] studies the under-parameterized setting with

multiple student and teacher neurons, but they relies on the correlation loss to decouple student neuron dynamics, potentially concealing important phenomena.

The under-parameterized teacher-student setup is directly related to distillation. In fact, it is equivalent to knowledge distillation in binary classification NNs [cite hinton]. Existing theories of NN distillation require the linear representation hypotheses [cite enric]. Additionally, prior works on the superposition of features study empirically study and interpret different under-parameterized models [cite anthropic].

1.2 Our model

We study the teacher-student setup and focus on the under-parameterized regime, since it is understudied and relevant to distillation, superposition, and large language model training. To be as realistic as possible, we consider the squared loss and multiple teacher and student neurons. To keep our analysis tractable, we consider the population spherical gradient flow dynamics with even Hermite polynomial activation functions for both the teacher and student and orthonormal teacher vectors. Even with these assumptions, it is difficult to get a precise understanding of the critical points, let alone the dynamics. Thus, we began by studying a toy model with 2 teacher neurons and 2 student neurons. Then, we generalized the results to the exact and under-parameterized settings.

We now formally introduce our model. We consider the following (typically non-convex) optimization problem

$$L^{n,k}(\{w_i\}_{i=1}^n) = \frac{1}{2} \mathop{\mathbb{E}}_{x\sim\mathcal{D}} \Big[\Big(\sum_{i=1}^n H_\rho(w_i^\top x) - \sum_{j=1}^k H_\rho(v_j^\top x) \Big)^2 \Big],$$

where $\mathcal{D} = \mathcal{N}(0, I_d)$, $w_i, v_j \in \mathbb{S}^{d-1}$, and H_ρ is the ρth normalized Hermite polynomial. We can expand out the square and use the linearity of expectation to rewrite our objective as

$$= \frac{1}{2} \sum_{i=1}^n \sum_{l\neq i}^n g_\rho(w_i^\top x, w_l^\top x) - \sum_{i=1}^n \sum_{j=1}^k g_\rho(w_i^\top x, v_j^\top x) + \frac{1}{2} \sum_{j=1}^k \sum_{j'=1}^k g_\rho(v_j^\top x, v_{j'}^\top x),$$

where $g_\rho(a,b) = \mathbb{E}[H_\rho(a)H_\rho(b)]$. Since w_i and v_j are unit vectors, the correlation between $w_i^\top x$ and $v_j^\top x$ is $w_i^\top v_j$. By Proposition 11.31 in O'Donnell (2021), $g_\rho(w_i^\top x, v_j^\top x) = (w_i^\top v_j)^\rho$. The first and last terms are constants in terms of $w_i \in \mathbb{S}^{d-1}$, because of the spherical constraint and independence of w_i respectively. We can rewrite our objective as

$$= \frac{1}{2} \sum_{i=1}^n \sum_{q=1}^n (w_i^\top w_q)^\rho - \sum_{i=1}^n \sum_{j=1}^k (w_i^\top v_j)^\rho + C = \sum_{i=1}^n \sum_{i'>i}^n (w_i^\top w_{i'})^\rho - \sum_{i=1}^n \sum_{j=1}^k (w_i^\top v_j)^\rho + C.$$

Setting

$$W = \begin{bmatrix} — & w_1^\top & — \\ & \vdots & \\ — & w_n^\top & — \end{bmatrix} \in \mathbb{R}^{n\times d} \text{ and } V = \begin{bmatrix} — & v_1^\top & — \\ & \vdots & \\ — & v_k^\top & — \end{bmatrix} \in \mathbb{R}^{k\times d}$$

and denoting $(\cdot)^{\circ\rho}$ as the Hadamard power that raises each element of its input to the ρth power, we can rewrite the objective as

$$= \frac{1}{2} \mathbb{1}_n^\top (WW^\top)^{\circ\rho} \mathbb{1}_n - \mathbb{1}_n^\top (WV^\top)^{\circ\rho} \mathbb{1}_k + C,$$

where $\mathbb{1}_n \in \{1\}^n$. This model is identical, up to constant factors, to that of [cite martin] when the activation function is the second normalized Hermite polynomial. See appendix A for more details.

1.3 Contributions

In section 2, we conduct a full critical point analysis of the toy model with two student and two teacher neurons. We also briefly describe the critical points in more general cases. In section 3, we discuss dynamics, and prove that we avoid "tight-balance" critical points from favorable initializations. We generalize these initializations to the multi teacher and student neuron exact and under-parameterized setting. In section 4, we connect our results to machine learning practice, in particular to activation function choices, distillation and feature superposition.

Full proofs are in corresponding appendices.

2 Critical point analysis

2.1 Derivative

We first compute the derivative of our objective with respect to $w_{\alpha,\beta}$,

$$\frac{\partial L^{n,k}(W)}{\partial w_{\alpha,\beta}} = \rho w_{:,\beta}^\top (W w_\alpha)^{\circ\rho-1} - \rho v_{:,\beta}^\top (V w_\alpha)^{\circ\rho-1}.$$

Therefore,

$$\frac{\partial L^{n,k}(W)}{\partial W} = \rho (W W^\top)^{\circ\rho-1} W - \rho (W V^\top)^{\circ\rho-1} V.$$

We now consider the spherical derivative. Specifically, we restrict the rows of the derivative to be on the sphere by projecting each row onto the tangent space $T_{w_\alpha}(\mathbb{S}^{d-1})$ of the sphere

$$\begin{aligned}\overline{\frac{\partial L^{n,k}(W)}{\partial W}} &= \begin{bmatrix} — & P_{w_1}\left(\frac{\partial L^{n,k}(W)}{\partial w_1}\right) & — \\ & \vdots & \\ — & P_{w_n}\left(\frac{\partial L^{n,k}(W)}{\partial w_n}\right) & — \end{bmatrix} \\ &= \rho (W W^\top)^{\circ\rho-1} W - \rho (W V^\top)^{\circ\rho-1} V \\ &\quad - \left(\left[\rho (W W^\top)^{\circ\rho-1} W - \rho (W V^\top)^{\circ\rho-1} V\right] W^\top \circ I_n\right) W.\end{aligned}$$

Let $A = \rho (W W^\top)^{\circ\rho-1} W - \rho (W V^\top)^{\circ\rho-1} V$, then

$$\overline{\frac{\partial L^{n,k}(W)}{\partial W}} = A - (A W^\top \circ I_n) W.$$

2.2 Finding critical points

We set the derivative to 0

$$B - (B W^\top \circ I_n) W = 0,$$

where $B = A/\rho$. We identify various critical points.

2.2.1 Orthogonal copy critical points

Theorem 2.1 (Orthogonal copy critical points)**.** *Every arrangement of student vectors that satisfies the following conditions is a critical point:*

- *for each $i \in [n]$, $w_i = v_j$ for some v_j,*
- *for all $i \in [n]$ and $j \in [k]$, $w_i^\top v_j \in \{-1, 0, 1\}$.*

Proof. Consider the case where ρ is odd. Then the ith row of B is

$$\begin{aligned} B_i &= \sum_{j=1}^{n} (w_j^\top w_i)^{\rho-1} w_j - \sum_{j=1}^{k} (v_j^\top w_i)^{\rho-1} v_j \\ &= w_i \sum_{j=1}^{n} w_j^\top w_i - w_i \sum_{j=1}^{k} v_j^\top w_i. \end{aligned}$$

Since $\|w_i\|_2^2 = 1$, the ith row of the spherical gradient simplifies to

$$\begin{aligned} \overline{\frac{\partial L^{n,k}(W)}{\partial w_i}} &= w_i \sum_{j=1}^{n} w_j^\top w_i - w_i \sum_{j=1}^{k} v_j^\top w_i - \Big(w_i \sum_{j=1}^{n} w_j^\top w_i - w_i \sum_{j=1}^{k} v_j^\top w_i \Big)^\top w_i w_i \\ &= w_i \sum_{j=1}^{n} w_j^\top w_i - w_i \sum_{j=1}^{k} v_j^\top w_i - w_i \sum_{j=1}^{n} w_j^\top w_i + w_i \sum_{j=1}^{k} v_j^\top w_i = 0. \end{aligned}$$

This completes the proof for the odd case.

Consider the case where ρ is even. Then the ith row of B is

$$\begin{aligned} B_i &= \sum_{j=1}^{n} (w_j^\top w_i)^{\rho-1} w_j - \sum_{j=1}^{k} (v_j^\top w_i)^{\rho-1} v_j \\ &= w_i \sum_{j=1}^{n} |w_j^\top w_i| - w_i \sum_{j=1}^{k} |v_j^\top w_i|. \end{aligned}$$

Since $\|w_i\|_2^2 = 1$, the ith row of the spherical gradient simplifies to

$$\begin{aligned} \overline{\frac{\partial L^{n,k}(W)}{\partial w_i}} &= w_i \sum_{j=1}^{n} |w_j^\top w_i| - w_i \sum_{j=1}^{k} |v_j^\top w_i| - \Big(w_i \sum_{j=1}^{n} |w_j^\top w_i| - w_i \sum_{j=1}^{k} |v_j^\top w_i| \Big)^\top w_i w_i \\ &= w_i \sum_{j=1}^{n} |w_j^\top w_i| - w_i \sum_{j=1}^{k} |v_j^\top w_i| - w_i \sum_{j=1}^{n} |w_j^\top w_i| + w_i \sum_{j=1}^{k} |v_j^\top w_i| = 0. \end{aligned}$$

This completes the proof for the even case. This completes the entire proof. □

2.2.2 Euclidean gradient can never be 0

We want to show that the Euclidean gradient can never be 0, i.e. $B \neq 0$.

Simple case $\rho = 2$. In the simple case $\Phi(W) = W$. A necessary condition for $W^\top W = V^\top V$, is

$$\mathrm{Tr}(W^\top W) = \mathrm{Tr}(V^\top V).$$

Observe

$$\mathrm{Tr}(W^\top W) = \mathrm{Tr}(WW^\top) = \|w_1\|^2 + \ldots + \|w_n\|^2 = n,$$

and similarly $\mathrm{Tr}(V^\top V) = k$. Therefore it is not true that $W^\top W = V^\top V$ for $V = n/k$.

Even ρ.

Lemma 2.2 (Maximum of sum of dot products)**.** *Suppose $v_1, \ldots, v_k \in \mathbb{R}^d$ are pairwise orthogonal and unit length. Consider the optimization problem*

$$\max_{w \in \mathbb{S}^{d-1}} \sum_{j=1}^{k} (w^\top v_j)^\rho.$$

The solution is 1. When $\rho = 2$, span$(v_1, \ldots, v_k) \cup \mathbb{S}^{d-1}$ is the set of all w that attain 1. When $\rho > 2$, $\{v_1, \ldots, v_k\}$ is the set of all w that attain 1.

Proof. When $\rho = 2$, observe

$$\sum_{j=1}^{k} (w^\top v_j)^2 = (Vw)^\top (Vw) = \|Vw\|_2^2.$$

Consider the orthonormal matrix $V' \in \mathbb{R}^{d \times d}$ which has $v_1, \ldots, v_k$ as its first k rows and other unit vectors for the last $d - k$ rows that satisfy orthogonality. Since $w \in \mathrm{span}(v_1, \ldots, v_k)$, the inner product of w and any of the last $d - k$ rows of V' is 0. Therefore $\|Vw\|_2^2 = \|V'w\|_2^2$. Observe V' is a rotation matrix and recall rotation matrices preserve distances. Therefore $\|V'w\|_2^2 = \|w\|_2^2 = 1$, proving the first case.

Consider $\rho > 2$. **BNOTE**: analysis copied from Berfin Let $u = Vw$. By the orthogonality of the teacher vectors, any $u \in \mathrm{span}(v_1, \ldots, v_k) \cup \mathbb{S}^{d-1}$ corresponds to a unique $w \in \mathrm{span}(v_1, \ldots, v_k)$. Therefore we can write the Lagrangian function as

$$\mathcal{L}(u) = \sum_{j=1}^{k} u_j^\rho + \lambda \Big(\sum_{j=1}^{k} u_j^2 - 1 \Big).$$

Recall,

$$\frac{d\mathcal{L}}{du_j} = \rho u_j^{\rho-1} + 2\lambda u_j = 0$$

is a condition for the maximum, and thus

$$\rho u_j^{\rho-2} = -2\lambda.$$

The left hand side is a strictly increasing function of u_j for all $\rho > 2$, therefore there is a unique u_j^* that satisfies this equation. This implies that all non-zero coordinates of u must be equal and that candidates for the optimal solution have the form

$$u^*(l) = \Big(\underbrace{0, \ldots, 0}_{k-l}, \underbrace{\frac{1}{\sqrt{l}}, \ldots, \frac{1}{\sqrt{l}}}_{l} \Big), \quad l \in [k].$$

Evaluating the objective for all these candidate solutions, we have

$$\sum_{j=1}^{k} u_j^{*\rho} = l^{1-\rho/2},$$

which is maximized to 1 at $l = 1$ for every $\rho > 2$, implying the optimal solution is $u^* = (1, 0, \ldots, 0)$. This corresponds to some v_j (or $-v_j$ in the case of even ρ), proving the second case. □

Lemma 2.3 (Orthogonal copy critical points)**.** *If the teacher vectors are orthogonal, ρ is even, $\rho > 0$, and $n < k$, there exist no configuration of student neurons such that $B = 0$.*

Proof. We aim to show $B \neq 0$. Observe a sufficient condition is that there exists a $i \in [n]$ such that the squared ℓ^2 norm of the ith row of $(WW^\top)^{\circ\rho-1}W$ is not equal to the squared ℓ^2 norm of the ith row of $(WV^\top)^{\circ\rho-1}V$.

We compute the squared ℓ^2 norm of the ith row of $(WV^\top)^{\circ\rho-1}V$,

$$\|((WV^\top)^{\circ\rho-1}V)_i\|_2^2 = \|\sum_{j=1}^{k}(v_j^\top w_i)^{\rho-1}v_j\|_2^2 = V\sum_{j=1}^{k}(v_j^\top w_i)^{2\rho-2}.$$

By lemma 2.2, we know this quantity is at most V, with equality only when $w_i = v_j$ for some $j \in [k]$.

We compute the squared ℓ^2 norm of the ith row of $(WW^\top)^{\circ\rho-1}W$,

$$\begin{aligned}
\|((WW^\top)^{\circ\rho-1}W)_i\|_2^2 &= \|\sum_{j=1}^{k}(w_j^\top w_i)^{\rho-1}w_j\|_2^2 \\
&= 2\sum_{j'>j}^{n}(w_j^\top w_i)^{\rho-1}(w_{j'}^\top w_i)^{\rho-1}w_{j'}^\top w_j + \sum_{j=1}^{n}(w_j^\top w_i)^{2\rho-2} \\
&= 2\sum_{j\neq i}^{n}(w_j^\top w_i)^{\rho} + 2\sum_{\substack{j'>j \\ j',j\neq i}}^{n}(w_j^\top w_i)^{\rho-1}(w_{j'}^\top w_i)^{\rho-1}w_{j'}^\top w_j + 1 + \sum_{j\neq i}^{n}(w_j^\top w_i)^{2\rho-2}.
\end{aligned}$$

We wish to show this quantity is at least 1, therefore showing that the norms can never be equal and proving the original statement. It is sufficient to show

$$2\sum_{\substack{j'>j \\ j',j\neq i}}^{n}(w_j^\top w_i)^{\rho-1}(w_{j'}^\top w_i)^{\rho-1}w_{j'}^\top w_j + \sum_{j\neq i}^{n}(w_j^\top w_i)^{2\rho-2} \geq 0.$$

Observe

$$2\sum_{\substack{j'>j \\ j',j\neq i}}^{n}(w_j^\top w_i)^{\rho-1}(w_{j'}^\top w_i)^{\rho-1}w_{j'}^\top w_j + \sum_{j\neq i}^{n}(w_j^\top w_i)^{2\rho-2} = \left\|\sum_{j\neq i}^{n}(w_j^\top w_i)^{\rho-1}w_j\right\|_2^2 \geq 0.$$

This completes the proof. □

2.2.3 Scaled rows critical points

Recall our critical point condition is

$$B - (BW^\top \circ I_n)W = 0.$$

This is equivalent to the condition for all $i \in [n]$, $B_i = (B_i^\top w_i)w_i$. Observe it suffices to say B_i is some scalar multiple of w_i since if there exists some constant $\beta_i \in \mathbb{R}$ such that $B_i = \beta_i w_i$,

$$\begin{aligned}\beta_i w_i &= (\beta_i w_i^\top w_i)w_i \\ \beta_i w_i &= \beta_i w_i,\end{aligned}$$

implying $\beta_i = B_i^\top w_i$.

Expanding out B_i, we have

$$\begin{aligned}B_i &= W^\top (Ww_i)^{\circ\rho-1} - V^\top (Vw_i)^{\circ\rho-1} \\ &= w_i + \sum_{j\neq i}(w_i^\top w_j)^{\rho-1}w_j - \sum_{j=1}^{k}(w_i^\top v_j)^{\rho-1}v_j.\end{aligned}$$

Student in span of teachers. We first consider the case where $w_i \in \operatorname{span}(\{v_1, \dots, v_k\})$. Since the teacher vectors are unit norm and orthogonal, there exists some $\alpha_i \in \mathbb{S}^{k-1}$ such that $w_i = \alpha_{i,1}v_1 + \cdots + \alpha_{i,k}v_k$. We must now show that there exists some $\beta_i \in \mathbb{R}$ such that

$$\sum_{j\neq i}^{n}(\alpha_i^\top \alpha_j)^{\rho-1}(\alpha_{j,1}v_1 + \cdots + \alpha_{j,k}v_k) - \sum_{j=1}^{k}\alpha_{i,j}^{\rho-1}v_j = (\beta_i - 1)(\alpha_{i,1}v_1 + \cdots + \alpha_{i,k}v_k).$$

Since the teacher vectors are orthogonal, an equivalent condition is for all $i \in [n]$ and for all $l \in [k]$,[1]

$$\sum_{j\neq i}^{n}(\alpha_i^\top \alpha_j)^{\rho-1}\alpha_{j,l} - \alpha_{i,l}^{\rho-1} = (\beta_i - 1)\alpha_{i,l}.$$

We can multiply each side of the expression above by $\alpha_{i,l}$ for all $l \in [k]$,

$$\sum_{j\neq i}^{n}(\alpha_i^\top \alpha_j)^{\rho-1}\alpha_{i,l}\alpha_{j,l} - \alpha_{i,l}^{\rho} = (\beta_i - 1)\alpha_{i,l}^2.$$

Summing these equations over all $l \in [k]$ for a fixed i,

$$\begin{aligned}\sum_{j\neq i}^{n}(\alpha_i^\top \alpha_j)^{\rho-1}\sum_{l=1}^{k}\alpha_{i,l}\alpha_{j,l} - \sum_{l=1}^{k}\alpha_{i,l}^{\rho} &= (\beta_i - 1)\sum_{l=1}^{k}\alpha_{i,l}^2 \\ \sum_{j\neq i}^{n}(\alpha_i^\top \alpha_j)^{\rho} - \|\alpha_i\|_\rho^\rho &= \beta_i - 1.\end{aligned}$$

[1] If the student vectors are orthogonal, the condition simplifies to $-\alpha_{i,l}^{\rho-1} = (\beta_i - 1)\alpha_{i,l}$, which clearly holds if $\alpha_{i,l} \in \{-1, 0, 1\}$. This is similar to a part of the proof of lemma 2.3.

Solving for β_i,

$$\beta_i = 1 + \sum_{j \neq i}^{n} (\alpha_i^\top \alpha_j)^\rho - \|\alpha_i\|_\rho^\rho.$$

Substituting β_i to our prior equation,

$$\sum_{j \neq i}^{n} (\alpha_i^\top \alpha_j)^{\rho-1} \alpha_{j,l} - \alpha_{i,l}^{\rho-1} = (\sum_{j \neq i}^{n} (\alpha_i^\top \alpha_j)^\rho - \|\alpha_i\|_\rho^\rho) \alpha_{i,l}.$$

Writing the equalities in a vectorized form, we have an equivalent critical points condition that is more interpretable,

$$\begin{aligned}
\sum_{j \neq i}^{n} (\alpha_i^\top \alpha_j)^{\rho-1} \alpha_j - \alpha_i^{\circ \rho-1} &= \sum_{j \neq i}^{n} (\alpha_i^\top \alpha_j)^\rho \alpha_i - \|\alpha_i\|_\rho^\rho \alpha_i \\
\sum_{j \neq i}^{n} (\alpha_i^\top \alpha_j)^{\rho-1} \Big(\alpha_j - (\alpha_i^\top \alpha_j) \alpha_i \Big) &= \Big(\alpha_i^{\circ \rho-1} - \|\alpha_i\|_\rho^\rho \alpha_i \Big) \\
\sum_{j \neq i}^{n} (\alpha_i^\top \alpha_j)^{\rho-1} \Big(\alpha_j - (\alpha_i^\top \alpha_j) \alpha_i \Big) &= \Big(\alpha_i^{\circ \rho-1} - (\alpha_i^{\circ \rho-1 \top} \alpha_i) \alpha_i \Big).
\end{aligned}$$

Observe for any vectors $u, v \in \mathbb{R}^d$, $u - (u^\top v) v$ is the orthogonal projection of u onto the orthogonal complement of v.

2.2.4 $n = k = d = 2$ critical points

In general, it is still difficult to find critical points from the simplified form in the previous section. Therefore we consider a simplified case where $n = k = d = 2$ and $\rho > 2$ and ρ is even. In this case, W is a critical point iff it satisfies

$$\begin{aligned}
(\alpha_1^\top \alpha_2)^{\rho-1} \Big(\alpha_2 - (\alpha_1^\top \alpha_2) \alpha_1 \Big) &= \alpha_1^{\circ \rho-1} - (\alpha_1^{\circ \rho-1 \top} \alpha_1) \alpha_1 \\
(\alpha_2^\top \alpha_1)^{\rho-1} \Big(\alpha_1 - (\alpha_2^\top \alpha_1) \alpha_2 \Big) &= \alpha_2^{\circ \rho-1} - (\alpha_2^{\circ \rho-1 \top} \alpha_2) \alpha_2,
\end{aligned}$$

where $\alpha_i = (w_i^\top v_1, w_i^\top v_2)$. We describe all ways this is possible,

Recall $\|\alpha_i\|_2 = 1$. Therefore, we can rewrite the conditions above in terms of α_{11} and α_{21} in non-vectorized form. For simplicity we denote $a = \alpha_{11}$, $b = \alpha_{21}$, and $c = ab + \sqrt{(1-a^2)(1-b^2)}$,

$$\begin{aligned}
c^{\rho-1} \Big(b - ca \Big) - a^{\rho-1} + (a^\rho + \sqrt{1-a^2}^\rho) a &= 0 \\
c^{\rho-1} \Big(\sqrt{1-b^2} - c\sqrt{1-a^2} \Big) - \sqrt{1-a^2}^{\rho-1} + (a^\rho + \sqrt{1-a^2}^\rho) \sqrt{1-a^2} &= 0 \\
c^{\rho-1} \Big(a - cb \Big) - b^{\rho-1} + (b^\rho + \sqrt{1-b^2}^\rho) b &= 0 \\
c^{\rho-1} \Big(\sqrt{1-a^2} - c\sqrt{1-b^2} \Big) - \sqrt{1-b^2}^{\rho-1} + (b^\rho + \sqrt{1-b^2}^\rho) \sqrt{1-b^2} &= 0.
\end{aligned}$$

Note it is difficult to find the zeros of these functions. Therefore, we plot these functions for $\rho = 4$ in figure 1 and by inspection find a full description of the critical points (all green dots in figure 1) in table 1.

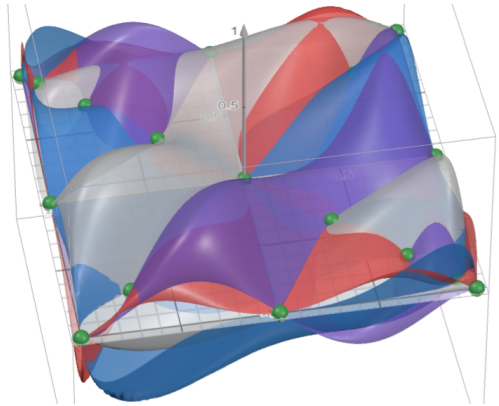

Figure 1: Critical points condition functions when $n = k = d = 2$, $\rho = 4$. All critical points are labeled with green dots.

Name	**Condition**
Copy	$w_1 = v_1$ and $w_2 = v_2$
Tight balance	$w_1^\top v_1 = w_2^\top v_1 = \cos^{-1}(\pi/8)$ and $w_1^\top v_2 = -w_2^\top v_2 = \sin^{-1}(\pi/8)$
Balance	$w_1^\top v_1 = w_2^\top v_1 = 2^{-1/2}$ and $w_1^\top v_1 = -w_2^\top v_1 = 2^{-1/2}$
Same copy	$w_1 = w_2 = v_1$
Same balance	$w_1 = w_2$, $w_1^\top v_1 = 2^{-1/2}$, and $w_1^\top v_2 = 2^{-1/2}$

Table 1: Full description of the critical points found from figure 1 up to sign and teacher symmetries.

When we vary ρ, each critical point stays the same except the tight balance critical point. Specifically, the angle of the student vectors to the closer teacher decreases as ρ increases. We show how to find this angle θ_ρ.

Observe in the tight balance critical point, $a = -b$ and consequentially $\sqrt{1-a^2} = \sqrt{1-b^2}$. In this case, $c = -a^2 + \sqrt{(1-a^2)^2} = 1 - 2a^2$ and we can simplify our critical point conditions to

$$(1-2a^2)^{\rho-1}\Big(-a-(1-2a^2)a\Big) - a^{\rho-1} + (a^\rho + \sqrt{1-a^2}^\rho)a = 0$$
$$(1-2a^2)^{\rho-1}\Big(\sqrt{1-a^2} - (1-2a^2)\sqrt{1-a^2}\Big) - \sqrt{1-a^2}^{\rho-1} + (a^\rho + \sqrt{1-a^2}^\rho)\sqrt{1-a^2} = 0.$$

We can further simplify to

$$(1-2a^2)^{\rho-1}(-2a+2a^3) - a^{\rho-1} + a^{\rho+1} - a\sqrt{1-a^2}^\rho = 0$$
$$2a^2\sqrt{1-a^2}(1-2a^2)^{\rho-1} - \sqrt{1-a^2}^{\rho-1} + a^\rho\sqrt{1-a^2} - \sqrt{1-a^2}^{\rho+1} = 0.$$

Recall that at a tight balance critical point, $a \neq 0$, therefore we can simplify the conditions to

$$-2(1-2a^2)^{\rho-1} - a^{\rho-2} + 2a^2(1-2a^2)^{\rho-1} + a^\rho - \sqrt{1-a^2}^\rho = 0$$
$$-\sqrt{1-a^2}^{\rho-2} + 2a^2(1-2a^2)^{\rho-1} + a^\rho - \sqrt{1-a^2}^\rho = 0$$

By subtracting the second equation from the first, we get a more general condition

$$-2(1-2a^2)^{\rho-1} - a^{\rho-2} + \sqrt{1-a^2}^{\rho-2} = 0.$$

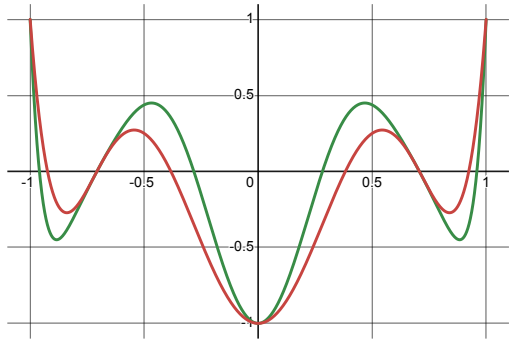

Figure 2: Tight balance critical point condition when $n = k = d = 2$. The red line corresponds to $\rho = 4$ and the green line corresponds to $\rho = 6$.

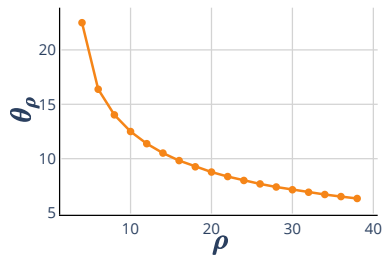

Figure 3: Tight balance critical point angle as a function of ρ.

Unfortunately, this condition is also difficult to solve analytically. We visualize the function in figure 2. As ρ varies between different even values larger than 2, the condition maintains the 0 at $2^{-1/2}$, but the other zeroes vary. We empirically solve for these zeros and plot them as a function of ρ in figure 3. For $\rho = 4$, $\theta_\rho = \pi/8$, and plugging this back into the conditions above verifies that it is a critical point. For larger ρ, we did not find a simple analytic form, yet empirically they satisfy the conditions above.

BNOTE: Show tight balance critical point in general proof

BNOTE: add pictures from the poster

2.3 Classifying critical points

Recall the spherical gradient is

$$\overline{\frac{\partial L^{n,k}(W)}{\partial W}} = A - (AW^\top \circ I_n)W,$$

where $A = \rho(WW^\top)^{\circ\rho-1}W - \rho(WV^\top)^{\circ\rho-1}V$.

The spherical gradient on the αth neuron is

$$\begin{aligned} &= \rho W^\top (Ww_\alpha)^{\circ\rho-1} - \rho w_\alpha w_\alpha^\top W^\top (Ww_\alpha)^{\circ\rho-1} \\ &\quad - \rho V^\top (Vw_\alpha)^{\circ\rho-1} + \rho w_\alpha w_\alpha^\top V^\top (Vw_\alpha)^{\circ\rho-1}. \end{aligned}$$

Let's assume $V = I_d = I_k$, then

$$\begin{aligned}&= \rho W^\top (W w_\alpha)^{\circ \rho - 1} - \rho w_\alpha w_\alpha^\top W^\top (W w_\alpha)^{\circ \rho - 1} \\ &\quad - \rho (w_\alpha)^{\circ \rho - 1} + \rho w_\alpha w_\alpha^\top (w_\alpha)^{\circ \rho - 1}.\end{aligned}$$

Do determine the sign of the gradient, we can divide by positive constants and let's assume $n = k$,

$$\begin{aligned}&= W^\top (W w_\alpha)^{\circ \rho - 1} - w_\alpha w_\alpha^\top W^\top (W w_\alpha)^{\circ \rho - 1} - \Big(w_\alpha^{\circ \rho - 1} - (w_\alpha^\top w_\alpha)^{\circ \rho - 1} w_\alpha \Big) \\ &= \sum_{j=1}^n (w_\alpha^\top w_j) w_j - w_\alpha w_\alpha^\top \sum_{j=1}^n (w_\alpha^\top w_j) w_j - \Big(w_\alpha^{\circ \rho - 1} - (w_\alpha^\top w_\alpha)^{\circ \rho - 1} w_\alpha \Big) \\ &= \sum_{j \neq \alpha} (w_\alpha^\top w_j)^{\circ \rho - 1} w_j - w_\alpha \sum_{j \neq \alpha} (w_\alpha^\top w_j)^{\circ \rho} - \Big(w_\alpha^{\circ \rho - 1} - (w_\alpha^\top w_\alpha)^{\circ \rho - 1} w_\alpha \Big)\end{aligned}$$

When $n = k = d = 2$,

$$= (w_1^\top w_2)^{\circ \rho - 1} \Big(w_2 - (w_1^\top w_2) w_1 \Big) - \Big(w_1^{\circ \rho - 1} - (w_1^\top w_1^{\circ \rho - 1}) w_1 \Big).$$

At the tight balance critical point this translates to

$$= (w_1^\top w_2)^{\circ \rho - 1} \Big(w_2 - (w_1^\top w_2) w_1 \Big) - \Big(w_1^{\circ \rho - 1} - (w_1^\top w_1^{\circ \rho - 1}) w_1 \Big).$$

From this form, by studying the signs you can get the dynamics.

Tight saddle can be shown by plugging in the condition?

Want to show saddles and local minima

Want to show

2.3.1 Lowest loss orthogonal copy

While all orthogonal copy student configurations are critical points of the spherical loss, certain configurations achieve lower loss. Recall the loss

$$L^{n,k}(\{w_i\}_{i=1}^n) = \frac{1}{2} \mathbb{1}_n^\top (W W^\top)^{\circ \rho} \mathbb{1}_n - \mathbb{1}_n^\top (W V^\top)^{\circ \rho} \mathbb{1}_k + C.$$

Through a series of short lemmas, we prove the optimal configuration in the under-parameterized regime with orthogonal teachers

Lemma 2.4 (Copying different neurons is better than copying the same)**.** *Assume the teacher vectors under consideration are orthogonal to all other teacher vectors. All else equal, two neurons copying different teachers has lower loss than two neurons copying the same teacher.*

Proof. Under the assumption, regardless of if two student vectors are copying the same teacher vector or different teachers vector, $WV^\top$ is the same. However, if two student vectors copy the same teacher vector, $WW^\top$ is larger than if two student vectors copy different teacher vectors. This completes the proof. □

Corollary 2.4.1 (Optimal orthogonal copy critical point)**.** *Assume the teacher vectors under consideration are orthogonal to all other teacher vectors. The orthogonal copy critical point with the lowest loss is if every student vector is equal to a different teacher vector.*

Proof. This follows directly from the previous lemma. □

2.3.2 Global minima

Theorem 2.5. *When the teacher vectors are pair-wise orthogonal and the student network is under or exactly parameterized and ρ is even, a student configuration where each student vector copies a different teacher vector (up to sign symmetries) achieves the lowest loss. When $\rho = 2$, any configuration where the students vector are orthogonal and lie in the span of the teacher vectors is also a global minimum.*

Proof. Recall the loss has two terms dependent on W, $\frac{1}{2}\,\mathbb{1}_n^\top (WW^\top)^{\circ\rho}\,\mathbb{1}_n$ and $-\,\mathbb{1}_n^\top (WV^\top)^{\circ\rho}\,\mathbb{1}_k$.

Consider the first term. Observe $(WW^\top)^{\circ\rho}$ always has ones on the main diagonal and non-negative off diagonal entries. Therefore, this term is minimized when the off diagonals are all 0, which is only the case when w_is are pairwise orthogonal.

Consider the second term. Observe $-\,\mathbb{1}_n^\top (WV^\top)^{\circ\rho}\,\mathbb{1}_k$ is minimized when $\sum_{j=1}^{k} (w_i^\top v_j)^\rho$ is maximized for each $i \in [n]$. By lemma 2.2, when $\rho > 2$ this is only the case if $w_i = v_j$ or -v_j. When $\rho = 2$, this is the case if w_i lies in the span of the teacher vectors.

Put together, this implies a configuration of student vectors is a global minimum iff it meets the minimum condition for both the first and the second term. The stated configurations are the only ones that achieve this. This completes the proof. □

3 Dynamics

We study the training dynamics under the spherical population gradient flow,

$$\frac{dW^{(t)}}{dt} = -\,\nabla^{\mathbb{S}^{d-1}} L^{n,k}(W) = -\overline{\frac{\partial L^{n,k}(W)}{\partial w_\alpha}}.$$

BNOTE: blah blah about what we are trying to do here.

3.1 Toy model cone initialization convergence

Lemma 3.1. *Suppose $n = k = d = 2$ and ρ is even and $\rho > 2$. If student vectors are initialized closest to different teacher vectors (up to sign symmetries), then the student vectors will converge to their closest teacher vectors.*

Proof Sketch. We first show if the student vectors are initialized closest to different teachers, they will remain closest to their respective teacher throughout all of training. We show this by studying the behavior of the spherical gradient flow when a student vector is nearly perfectly in between the two teacher vectors.

The only critical point attainable following split teacher initialization is the copy global minimum. Therefore gradient flow will converge to this critical point. □

Unfortunately, this split cone behavior is only exhibited when $d = 2$. The simplest counter example can be constructed when $n = 2$ and $k = d = 3$, where regardless of how we set V or if we use some other scaling, we are not guaranteed to stay in the cone we initialize in. However, with the right scaling in our objective, we can describe an slightly more restricted initialization from which we will converge to the globally optimal copy configuration.

Lemma 3.2. *Suppose $n \leq k \leq d < \infty$ and ρ is even and $\rho > 2$. If all student vectors are initialized such that $w_i^\top v_i > 2^{-1/2} + \epsilon$ for $\epsilon = 0.17$* ***BNOTE****: I think ϵ should be a function of ρ* ***BNOTE****: say it in terms of angle too maybe, then the student vectors will converge to their closest teacher vectors.*

Proof. Proof sketch: first write out $f(\epsilon)$, then lower bound each of the terms and show that for some $\epsilon(\rho)$ it is greater than 0. Use the weird bounds I did and leave conjectures from desmos where needed.

We first show if the student vectors are initialized closest to different teachers, they will remain closest to their respective teacher throughout all of training. We show this by showing spherical gradient flow takes a student vector towards the closest teacher if the student vector is within a certain distance of the teacher.

WLOG we assume $v_i = e_i$ **BNOTE**: elaborate why this is ok!.

Then the spherical gradient on the ith student vector is

$$\begin{aligned}
\overline{\frac{\partial L^{n,k}(W)}{\partial w_i}} &= \rho\Bigg(\sum_{j\neq i}(w_i^\top w_j)^{\rho-1}(w_j - (w_i^\top w_j)w_i) - \sum_{j=1}^{k}(w_{i,j})^{\rho-1}(e_j - w_{i,j}w_i)\Bigg) \\
&= \rho\Bigg(\sum_{j\neq i}(w_i^\top w_j)^{\rho-1}(w_j - (w_i^\top w_j)w_i) - (w_i^{\circ\rho-1} - w_i\sum_{j=1}^{k} w_{i,j}^{\rho})\Bigg).
\end{aligned}$$

For convenience, we omit the multiplicative ρ factor and consider how much the spherical gradient flow points in the direction of the closest teacher vector $v_1 - (w_i^\top v_1)w_i = e_i - w_{i,i}w_i$, **BNOTE**: need to carefully argue this part tbh

$$\begin{aligned}
&-\overline{\frac{\partial L^{n,k}(W)}{\partial w_i}}^\top (e_i - w_{i,i}w_i) \\
&\qquad = -\Bigg(\sum_{j\neq i}(w_i^\top w_j)^{\rho-1}(w_j - (w_i^\top w_j)w_i) - (w_i^{\circ\rho-1} - w_i\sum_{j=1}^{k} w_{i,j}^{\rho}w_i)\Bigg)^\top (e_i - w_{i,i}w_i) \\
&\qquad = -\Bigg(\sum_{j\neq i}(w_i^\top w_j)^{\rho-1}(w_j - (w_i^\top w_j)w_i) - (w_i^{\circ\rho-1} - w_i\sum_{j=1}^{k} w_{i,j}^{\rho})\Bigg)^\top (e_i) \\
&\qquad = (w_{i,i}^{\circ\rho-1} - w_{i,i}\sum_{j=1}^{k} w_{i,j}^{\rho}) - \sum_{j\neq i}(w_i^\top w_j)^{\rho-1}(w_{j,i} - (w_i^\top w_j)w_{i,i}).
\end{aligned}$$

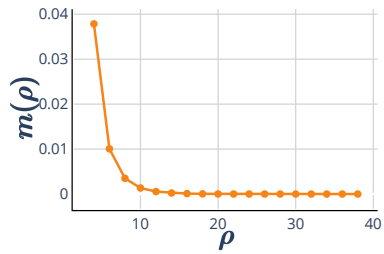

Figure 4: Lower bound on ϵ with nice properties.

Suppose $w_{i,i} = 2^{-1/2} + \epsilon$,

$$
\begin{aligned}
f(\epsilon) &= ((2^{-1/2}+\epsilon)^{\circ\rho-1} - (2^{-1/2}+\epsilon)\sum_{j=1}^{k} w_{i,j}^{\rho}) - \sum_{j\neq i}(w_i^\top w_j)^{\rho-1}(w_{j,i} - (w_i^\top w_j)(2^{-1/2}+\epsilon)) \\
&= (2^{-1/2}+\epsilon)^{\circ\rho-1} - (2^{-1/2}+\epsilon)^{\rho+1} - (2^{-1/2}+\epsilon)\sum_{j\neq i}^{k} w_{i,j}^{\rho} \\
&\quad - \sum_{j\neq i}^{n}(w_i^\top w_j)^{\rho-1} w_{j,i} + (2^{-1/2}+\epsilon)\sum_{j\neq i}^{n}(w_i^\top w_j)^{\rho}.
\end{aligned}
$$

The first three terms represent the "attractive" student to teacher force, and the last two terms represent the "repulsive" student to student force. By lemma 2.2, the attractive terms are minimized when there exists some $l \in [k] \setminus i$ such that $w_{i,l} = \sqrt{1-(2^{-1/2}+\epsilon)^2}$ and all other entries of w_i are 0. We write this formally as

$$
a(\epsilon) \geq (2^{-1/2}+\epsilon)^{\circ\rho-1} - (2^{-1/2}+\epsilon)^{\rho+1} - (2^{-1/2}+\epsilon)\sqrt{1-(2^{-1/2}+\epsilon)^2}^{\rho}
$$

The repulsive terms can be lower bounded as

$$
\begin{aligned}
r(\epsilon) &\geq -(2^{-1/2}+\epsilon)\Big(\sum_{j\neq i}^{n}(w_i^\top w_j)^{\rho-1} - \sum_{j\neq i}^{n}(w_i^\top w_j)^{\rho}\Big) \\
&\geq \min_{\theta\in[0,1-(2^{-1/2}+\epsilon)^2]} -(2^{-1/2}+\epsilon)(n-1)(\theta^{\rho-1}-\theta^{\rho}) \\
&= -(2^{-1/2}+\epsilon)(n-1)\Big((1-(2^{-1/2}+\epsilon)^2)^{\rho-1} - (1-(2^{-1/2}+\epsilon)^2)^{\rho}\Big)
\end{aligned}
$$

BNOTE: prove the last equality since $w_{j,i} \leq \sqrt{1-(2^{-1/2}+\epsilon)^2} \leq (2^{-1/2}+\epsilon)$.

We find for $\epsilon \geq m(\rho)$, with $1/n$ scaling,

$$
f(\epsilon) = a(\epsilon) + \frac{1}{n} r(\epsilon) \geq 0,
$$

where $m(\rho) < \big(40(x-3.9)\big)^{-1}$, see figure 4.

□

4 Broader connections

4.1 Distillation

4.2 Activation funtion choices

Why don't people use even activation functions? They might run into tight balance critical points! Also, even activation functions with small derivative near 0 might lead to regions of very small gradients around initialization or when the magnitudes of weights are small or something.

We ran very simple experiments to test this and found that absolute value activation functions actually work pretty well and functions with small derivatives near 0 tend to do poorly. Also activation functions that blow up more like quartic were very unstable in training actually. See figures 5 and 6.

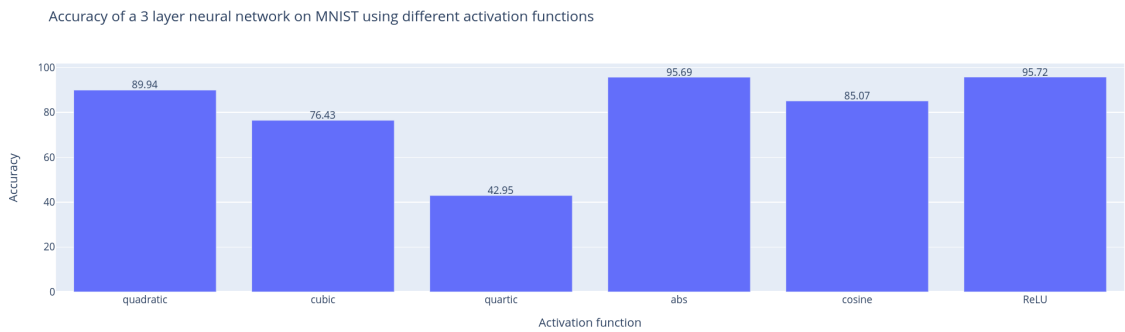

Figure 5: Performance of various even activation functions and ReLU on MNIST. Trained a 3 layer neural network with 100 neurons per layer. Trained with Adam.

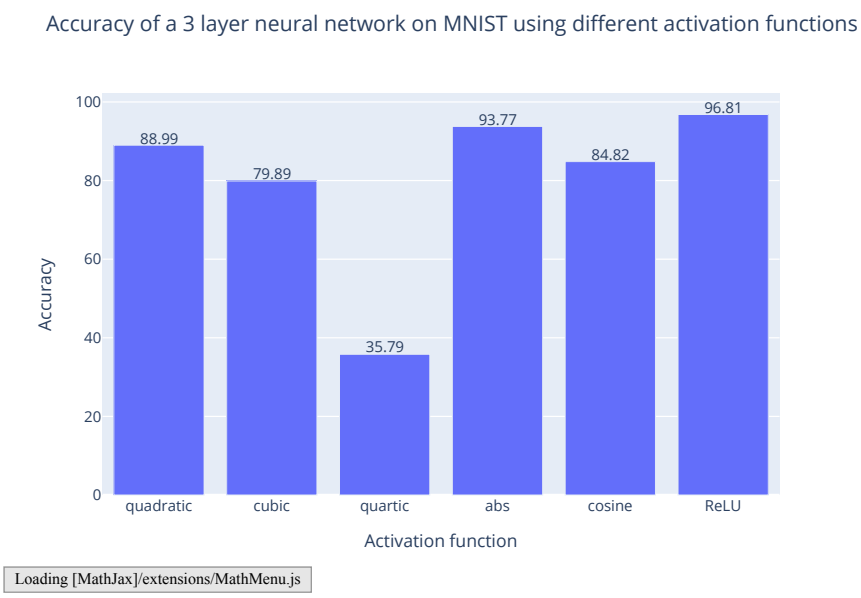

Figure 6: Same as above just retrained.

5 Discussion and future directions

We could have also taken limits instead of doing the toy model. There's no good reason we didn't do this, it could be cool future work!

BNOTE: need to update

It's time to think long term.

Things for write-up:

- Contrast the quadratic case with our case.
- Proof of cone initialization in $n = k = 2$.
- Proof of critical points in $n = k = 2$.
- Proof of saddles / minima in $n = k = 2$.
- Proof of global minimum in all under/exactly parameterized cases.
- Add plots and analysis of plots.
- Get better plot for initial training following correlation loss, then finding saddle, and finally converging to copy for $n = k = 2$, $d = 3$.
- Clean up critical points in general the under/exactly parameterized cases and add case where vectors are equal up to signs.

Things for later (in order of priority):

1. See if we can find nice form for the critical point, e.g. $\pi^{\rho-3}/2^{\rho-1}$.
2. More general activation functions, e.g. general even, linear + even, and odd.
3. Generalize tight balance to general case.
4. Consider loss landscape in non-orthogonal case, is this akin to the non-isotropic case? Can we actually write the non-isotropic case as a non-orthogonal case? are they equivalent?
5. Convergence rates for $n = k = 2$.
6. Maybe add second layer weights on the student...

---

7. See if there is some sort of local convexity so that we can prove convergence from good initialization.
8. Prove under what initializations we reach the tight balance critical point.
9. Global convergence result, i.e. we will provably converge to Copy (might be hard).
10. See if there are more critical points in the general case, e.g. triple balance points.
11. Talk about early stage training approximating correlation until you hit the saddle in the span of the teachers.
12. Take some limits/mean-field to prove global convergence or smth in those cases.
13. Going beyond population gradient and analyzing sample complexity of gradient descent.

Possible TODOs:

- Formulate gradient flow on sphere
- Solve ODE for gradient flow with some initial condition W_0
- Determine convergence rates
- Find critical points of the objective
- Consider large d limit
- Consider large d, n, and k limit (mean-field limit right?)
- Simulate and see what happens

References

Ryan O'Donnell. Analysis of boolean functions, 2021.

A Relation to previous works with the quadratic activation function

We consider the similarity of our work to Proposition 4.1 in **?**. We now assume $w_i, v_j \in \mathbb{R}^d$. Recall

$$H_2(x) = \frac{1}{\sqrt{2}}(x^2 - 1).$$

By Isserlis' Theorem,

$$\begin{aligned}
g_2(w_i, v_j) &= \mathop{\mathbb{E}}_{x\sim\mathcal{D}}[H_2(w_i^\top x)H_2(v_j^\top x)] = \mathop{\mathbb{E}}_{a,b\sim\mathcal{N}(0,\Sigma)}[H_2(a)H_2(b)] \\
&= \frac{1}{2}\mathop{\mathbb{E}}_{a,b\sim\mathcal{N}(0,\Sigma)}[(a^2-1)(b^2-1)] \\
&= \frac{1}{2}\mathop{\mathbb{E}}_{a,b\sim\mathcal{N}(0,\Sigma)}[a^2b^2 - a^2 - b^2 + 1] \\
&= \frac{1}{2}\mathop{\mathbb{E}}_{a,b\sim\mathcal{N}(0,\Sigma)}[a^2b^2] - \frac{1}{2}\|w_i\|^2 - \frac{1}{2}\|v_j\|^2 + \frac{1}{2} \\
&= \frac{1}{2}\|w_i\|^2\|v_j\|^2 + (w_i^\top v_j)^2 - \frac{1}{2}\|w_i\|^2 - \frac{1}{2}\|v_j\|^2 + \frac{1}{2},
\end{aligned}$$

where

$$\Sigma = \begin{bmatrix} \|w_i\|^2 & w_i^\top v_j \\ w_i^\top v_j & \|v_j\|^2 \end{bmatrix}.$$

Clearly, if we keep the original assumption that $w_i, v_j \in \mathbb{S}^{d-1}$, we are left with $(w_i^\top v_j)^2$. On the other hand, **?** considers the squared loss, where,

$$\mathop{\mathbb{E}}_{x\sim\mathcal{D}}[(w_i^\top x)^2(v_j^\top x)^2] = \mathop{\mathbb{E}}_{a,b\sim\mathcal{N}(0,\Sigma)}[a^2b^2] = \|w_i\|^2\|v_j\|^2 + 2(w_i^\top v_j)^2.$$

Rewriting the loss for our setting with normalization,

$$
\begin{aligned}
L^{n,k}(\{w_i\}_{i=1}^n) &= \mathop{\mathbb{E}}_{x\sim\mathcal{D}}\Big[\Big(\frac{1}{n}\sum_{i=1}^n H_2(w_i^\top x) - \frac{1}{k}\sum_{j=1}^k H_2(v_j^\top x)\Big)^2\Big] \\
&= \frac{1}{n^2}\sum_{i=1}^n\sum_{q=1}^n g_\rho(w_i^\top x, w_q^\top x) - 2\frac{1}{nk}\sum_{i=1}^n\sum_{j=1}^k g_\rho(w_i^\top x, v_j^\top x) + \frac{V}{k^2}\sum_{j=1}^k\sum_{q=1}^k g_\rho(v_j^\top x, v_q^\top x) \\
&= \frac{1}{n^2}\sum_{i=1}^n\sum_{q=1}^n \frac{1}{2}\|w_i\|^2\|w_q\|^2 + (w_i^\top w_q)^2 - \frac{1}{2}\|w_i\|^2 - \frac{1}{2}\|w_q\|^2 + \frac{1}{2} \\
&\quad - 2\frac{1}{nk}\sum_{i=1}^n\sum_{j=1}^k \frac{1}{2}\|w_i\|^2\|v_j\|^2 + (w_i^\top v_j)^2 - \frac{1}{2}\|w_i\|^2 - \frac{1}{2}\|v_j\|^2 + \frac{1}{2} \\
&\quad + \frac{V}{k^2}\sum_{j=1}^k\sum_{r=1}^k \frac{1}{2}\|v_j\|^2\|v_r\|^2 + (v_j^\top v_r)^2 - \frac{1}{2}\|v_j\|^2 - \frac{1}{2}\|v_r\|^2 + \frac{1}{2} \\
&= \frac{1}{n^2}\sum_{i=1}^n\sum_{q=1}^n -\frac{1}{2}\|w_i\|^2 - \frac{1}{2}\|w_q\|^2 + \frac{1}{2} - 2\frac{1}{nk}\sum_{i=1}^n\sum_{j=1}^k -\frac{1}{2}\|w_i\|^2 - \frac{1}{2}\|v_j\|^2 + \frac{1}{2} \\
&\quad + \frac{V}{k^2}\sum_{j=1}^k\sum_{r=1}^k -\frac{1}{2}\|v_j\|^2 - \frac{1}{2}\|v_r\|^2 + \frac{1}{2} \\
&\quad + \frac{1}{2}\Bigg(\frac{1}{n^2}\sum_{i=1}^n\sum_{q=1}^n \|w_i\|^2\|w_q\|^2 + 2(w_i^\top w_q)^2 - 2\frac{1}{nk}\sum_{i=1}^n\sum_{j=1}^k \|w_i\|^2\|v_j\|^2 + 2(w_i^\top v_j)^2 \\
&\quad + \frac{V}{k^2}\sum_{j=1}^k\sum_{r=1}^k \|v_j\|^2\|v_r\|^2 + 2(v_j^\top v_r)^2\Bigg).
\end{aligned}
$$

Observe the term with the parentheses can be rewritten in terms of an expectation of the quadratic loss and the other terms can be simplified,

$$
\begin{aligned}
&= \frac{1}{2} + +V - \frac{1}{n}\sum_{i=1}^n \|w_i\|^2 + \frac{1}{n}\sum_{i=1}^n \|w_i\|^2 + \frac{1}{k}\sum_{j=1}^k \|v_j\|^2 - \frac{V}{k}\sum_{j=1}^k \|v_j\|^2 \\
&\quad + \frac{1}{2}\mathop{\mathbb{E}}_{x\sim\mathcal{D}}\Big[\Big(\frac{1}{n}\sum_{i=1}^n (w_i^\top x)^2 - \frac{1}{k}\sum_{j=1}^k (v_j^\top x)^2\Big)^2\Big].
\end{aligned}
$$

If we set $V = 1$, the derivative of our loss will be the same as **?** (up to another 1/2 factor in front of the expectation),

$$
= \frac{5}{2} + \frac{1}{2}\mathop{\mathbb{E}}_{x\sim\mathcal{D}}\Big[\Big(\frac{1}{n}\sum_{i=1}^n (w_i^\top x)^2 - \frac{1}{k}\sum_{j=1}^k (v_j^\top x)^2\Big)^2\Big].
$$

B Critical point analysis proofs

B.1 Derivative

We first compute the derivative of our objective with respect to $w_{\alpha,\beta}$,

$$\begin{aligned}
\frac{\partial L^{n,k}(W)}{\partial w_{\alpha,\beta}} &= \frac{1}{2}\frac{\partial}{\partial w_{\alpha,\beta}}\sum_{i=1}^{n}\sum_{q=1}^{n}(w_i^\top w_q)^\rho - \frac{\partial}{\partial w_{\alpha,\beta}}\sum_{i=1}^{n}\sum_{j=1}^{k}(w_i^\top v_j)^\rho \\
&= \sum_{i\in[n],i\neq\alpha}\frac{\partial}{\partial w_{\alpha,\beta}}(w_\alpha^\top w_i)^\rho + \frac{1}{2}\frac{\partial}{\partial w_{\alpha,\beta}}(w_\alpha^\top w_\alpha)^\rho - \sum_{j=1}^{k}\frac{\partial}{\partial w_{\alpha,\beta}}(w_\alpha^\top v_j)^\rho \\
&= \sum_{i\in[n],i\neq\alpha}\rho w_{i,\beta}(w_\alpha^\top w_i)^{\rho-1} + \rho w_{\alpha,\beta}(w_\alpha^\top w_\alpha)^{\rho-1} - \sum_{j=1}^{k}\rho v_{j,\beta}(w_\alpha^\top v_j)^{\rho-1} \\
&= \rho w_{:,\beta}^\top (Ww_\alpha)^{\circ\rho-1} - \rho v_{:,\beta}^\top (Vw_\alpha)^{\circ\rho-1}.
\end{aligned}$$

Therefore,

$$\frac{\partial L^{n,k}(W)}{\partial w_\alpha} = \rho W^\top (Ww_\alpha)^{\circ\rho-1} - \rho V^\top (Vw_\alpha)^{\circ\rho-1},$$

and

$$\begin{aligned}
\frac{\partial L^{n,k}(W)}{\partial W} &= \begin{bmatrix} \frac{\partial L^{n,k}(W)}{\partial w_{1,1}} & \\ & \ddots \end{bmatrix} \\
&= \rho(WW^\top)^{\circ\rho-1}W - \rho(WV^\top)^{\circ\rho-1}V.
\end{aligned}$$

B.2 Derivative on the sphere

We restrict the rows of the derivative to be on the sphere by projecting each row onto the tangent space $T_{w_\alpha}(\mathbb{S}^{d-1})$ of the sphere

$$\begin{aligned}
\overline{\frac{\partial L^{n,k}(W)}{\partial W}} &= \begin{bmatrix} — & P_{w_1}\left(\frac{\partial L^{n,k}(W)}{\partial w_1}\right) & — \\ & \vdots & \\ — & P_{w_n}\left(\frac{\partial L^{n,k}(W)}{\partial w_n}\right) & — \end{bmatrix} \\
&= \begin{bmatrix} — & (I - w_1 w_1^\top)\left(\frac{\partial L^{n,k}(W)}{\partial w_1}\right) & — \\ & \vdots & \\ — & (I - w_n w_n^\top)\left(\frac{\partial L^{n,k}(W)}{\partial w_n}\right) & — \end{bmatrix}.
\end{aligned}$$

Let $\mathrm{diag}(u)$ denote the matrix with the entries of the vector u as its diagonal. We consider the αth row of the projected derivative matrix

$$\begin{aligned}
\overline{\frac{\partial L^{n,k}(W)}{\partial w_\alpha}} &= (I - w_\alpha w_\alpha^\top)\Big(\rho W^\top (W w_\alpha)^{\circ \rho-1} - \rho V^\top (V w_\alpha)^{\circ \rho-1}\Big) \\
&= \rho W^\top (W w_\alpha)^{\circ \rho-1} - \rho w_\alpha w_\alpha^\top W^\top (W w_\alpha)^{\circ \rho-1} \\
&\quad - \rho V^\top (V w_\alpha)^{\circ \rho-1} + \rho w_\alpha w_\alpha^\top V^\top (V w_\alpha)^{\circ \rho-1} \\
&= \rho W^\top (W w_\alpha)^{\circ \rho-1} - \rho w_\alpha w_\alpha^\top W^\top (W w_\alpha)^{\circ \rho-1} \\
&\quad - \rho V^\top (V w_\alpha)^{\circ \rho-1} + \rho w_\alpha w_\alpha^\top V^\top (V w_\alpha)^{\circ \rho-1} \\
&= \rho W^\top (W w_\alpha)^{\circ \rho-1} - \rho w_\alpha w_\alpha^\top W^\top (W w_\alpha)^{\circ \rho-1} \\
&\quad - \rho V^\top (V w_\alpha)^{\circ \rho-1} + \rho w_\alpha w_\alpha^\top V^\top (V w_\alpha)^{\circ \rho-1}.
\end{aligned}$$

Putting it all together, our spherical gradient is

$$\begin{aligned}
\overline{\frac{\partial L^{n,k}(W)}{\partial W}} &= \rho (W W^\top)^{\circ \rho-1} W - \rho\big((W W^\top)^{\circ \rho-1} W W^\top \circ I_n\big) W \\
&\quad - \rho (W V^\top)^{\circ \rho-1} V + \rho\big((W V^\top)^{\circ \rho-1} V W^\top \circ I_n\big) W \\
&= \rho (W W^\top)^{\circ \rho-1} W - \rho (W V^\top)^{\circ \rho-1} V \\
&\quad - \Big(\Big[\rho (W W^\top)^{\circ \rho-1} W - \rho (W V^\top)^{\circ \rho-1} V\Big] W^\top \circ I_n\Big) W.
\end{aligned}$$

Let $A = \rho (W W^\top)^{\circ \rho-1} W - \rho (W V^\top)^{\circ \rho-1} V$, then

$$\overline{\frac{\partial L^{n,k}(W)}{\partial W}} = A - (A W^\top \circ I_n) W.$$

I have experimentally verified that this is the correct expression for the spherical gradient.

C Dynamics proofs

C.1 Toy model cone initialization convergence proof

Lemma C.1. *Suppose $n = k = d = 2$ and ρ is even and $\rho > 2$. If student vectors are initialized closest to different teacher vectors (up to sign symmetries), then the student vectors will converge to their closest teacher vectors.*

Proof. We first show if the student vectors are initialized closest to different teachers, they will remain closest to their respective teacher throughout all of training. We show this by studying the behavior of the spherical gradient flow when a student vector is nearly perfectly in between the two teacher vectors.

Recall the spherical gradient on the ith student vector is

$$\begin{aligned}
\overline{\frac{\partial L^{n,k}(W)}{\partial w_i}} &= \rho W^\top (Ww_i)^{\circ\rho-1} - \rho V^\top (Vw_i)^{\circ\rho-1} \\
&\quad - \rho w_i w_i^\top W^\top (Ww_i)^{\circ\rho-1} + \rho w_i w_i^\top V^\top (Vw_i)^{\circ\rho-1} \\
&= \rho\Bigg(w_i + \sum_{j\neq i} (w_i^\top w_j)^{\rho-1} w_j - \sum_{j=1}^{k} (w_i^\top v_j)^{\rho-1} v_j \\
&\quad - w_i w_i^\top w_i - w_i w_i^\top \sum_{j\neq i} (w_i^\top w_j)^{\rho-1} w_j + w_i w_i^\top \sum_{j=1}^{k} (w_i^\top v_j)^{\rho-1} v_j \Bigg) \\
&= \rho\Bigg(\sum_{j\neq i} (w_i^\top w_j)^{\rho-1} w_j - \sum_{j=1}^{k} (w_i^\top v_j)^{\rho-1} v_j - w_i \sum_{j\neq i} (w_i^\top w_j)^{\rho} + w_i \sum_{j=1}^{k} (w_i^\top v_j)^{\rho} \Bigg) \\
&= \rho\Bigg(\sum_{j\neq i} (w_i^\top w_j)^{\rho-1} (w_j - (w_i^\top w_j) w_i) - \sum_{j=1}^{k} (w_i^\top v_j)^{\rho-1} (v_j - (w_i^\top v_j) w_i) \Bigg).
\end{aligned}$$

We can split the gradient flow update into the sum of three terms, each of which lie in the tangent space of w_i,

$$-\underbrace{\rho (w_i^\top w_j)^{\rho-1} (w_j - (w_i^\top w_j) w_i)}_{1} + \underbrace{\rho (w_i^\top v_1)^{\rho-1} (v_1 - (w_i^\top v_1) w_i)}_{2} + \underbrace{\rho (w_i^\top v_2)^{\rho-1} (v_2 - (w_i^\top v_2) w_i)}_{3}.$$

We assume $w_i^\top v_i = 2^{-1/2} + \epsilon$ for small $\epsilon > 0$ and symmetrically $i \in [2]$. Observe that the update is only one dimensional, so as long as it is pointing in the direction of v_1, we will never escape the cone. Therefore we must only compare the magnitudes. Observe

$$\begin{aligned}
\left\| (w_i^\top v_1)^{\rho-1} (v_1 - (w_i^\top v_1) w_i) \right\|_2 &= (w_i^\top v_1)^{\rho-1} \left\| (v_1 - (w_i^\top v_1) w_i) \right\|_2 \\
&= (w_i^\top v_1)^{\rho-1} \sqrt{\|v_1\|^2 - 2(w_i^\top v_1)^2 + (w_i^\top v_1)^2 \|w_1\|} \\
&= (w_i^\top v_1)^{\rho-1} \sqrt{1 - (w_i^\top v_1)^2} = (w_i^\top v_1)^{\rho-1} (w_i^\top v_2).
\end{aligned}$$

We consider the case where $2^{-1/2} < w_2^\top v_2 < 2^{-1/2} + \epsilon$ and $w_2^\top v_1 < 0$, since otherwise, term 1 pushes w_1 in the direction of v_1.

We must show

$$-(w_1^\top w_2)^{\rho-1} \sqrt{1 - (w_1^\top w_2)^2} + (w_1^\top v_1)^{\rho-1} (w_1^\top v_2) - (w_1^\top v_2)^{\rho-1} (w_1^\top v_1) > 0.$$

Since $w_1^\top v_1 = 2^{-1/2} + \epsilon$,

$$w_1^\top v_2 = \sqrt{1 - (2^{-1/2} + \epsilon)^2},$$

and

$$-2^{-1/2}(2^{-1/2} + \epsilon) + 2^{-1/2} \sqrt{1 - (2^{-1/2} + \epsilon)^2} < w_1^\top w_2 < 0.$$

Therefore,

$$
\begin{aligned}
&(w_1^\top w_2)^{\rho-1}\sqrt{1-(w_1^\top w_2)^2} + (w_1^\top v_1)^{\rho-1}(w_1^\top v_2) - (w_1^\top v_2)^{\rho-1}(w_1^\top v_1) \\
&\qquad > (w_1^\top w_2)^{\rho-1} + (w_1^\top v_1)^{\rho-1}(w_1^\top v_2) - (w_1^\top v_2)^{\rho-1}(w_1^\top v_1) \\
&\qquad > \Big(-2^{-1/2}(2^{-1/2}+\epsilon) + 2^{-1/2}\sqrt{1-(2^{-1/2}+\epsilon)^2}\Big)^{\rho-1} \\
&\qquad\quad + (2^{-1/2}+\epsilon)^{\rho-1}\sqrt{1-(2^{-1/2}+\epsilon)^2} - \sqrt{1-(2^{-1/2}+\epsilon)^2}^{\rho-1}(2^{-1/2}+\epsilon) = f(\epsilon).
\end{aligned}
$$

When $\epsilon = 0$, $\sqrt{1-(2^{-1/2}+\epsilon)^2} = 2^{-1/2}$, and

$$
\begin{aligned}
f(0) &= \Big(-2^{-1/2}(2^{-1/2}+\epsilon) + 2^{-1/2}\sqrt{1-(2^{-1/2}+\epsilon)^2}\Big)^{\rho-1} \\
&\quad + (2^{-1/2}+\epsilon)^{\rho-1}\sqrt{1-(2^{-1/2}+\epsilon)^2} - \sqrt{1-(2^{-1/2}+\epsilon)^2}^{\rho-1}(2^{-1/2}+\epsilon) \\
&= \Big(-2^{-1/2}2^{-1/2} + 2^{-1/2}2^{-1/2}\Big)^{\rho-1} + (2^{-1/2})^{\rho-1}2^{-1/2} - (2^{-1/2})^{\rho-1}2^{-1/2} = 0.
\end{aligned}
$$

Additionally, the derivative with respect to ϵ is

$$
\begin{aligned}
f'(\epsilon) &= \Big(-2^{-1/2}\epsilon - \frac{2^{-1/2}+\epsilon}{2^{-1/2}\sqrt{1-(2^{-1/2}+\epsilon)^2}}\Big)\Big(-2^{-1/2}(2^{-1/2}+\epsilon) + 2^{-1/2}\sqrt{1-(2^{-1/2}+\epsilon)^2}\Big)^{\rho-2} \\
&\quad + (\rho-1)(2^{-1/2}+\epsilon)^{\rho-2}\sqrt{1-(2^{-1/2}+\epsilon)^2} + \frac{(2^{-1/2}+\epsilon)^{\rho}}{\sqrt{1-(2^{-1/2}+\epsilon)^2}} \\
&\quad - \sqrt{1-(2^{-1/2}+\epsilon)^2}^{\rho-1} + (\rho-1)(2^{-1/2}+\epsilon)\sqrt{1-(2^{-1/2}+\epsilon)^2}^{\rho-3}(2^{-1/2}+\epsilon).
\end{aligned}
$$

When $\epsilon = 0$, the derivative is equal to

$$
\begin{aligned}
f'(0) &= \Big(-\frac{2^{-1/2}}{2^{-1/2}2^{-1/2}}\Big)\Big(-2^{-1/2}(2^{-1/2}) + 2^{-1/2}2^{-1/2}\Big)^{\rho-2} + (\rho-1)(2^{-1/2})^{\rho-2}2^{-1/2} \\
&\quad + \frac{(2^{-1/2})^{\rho}}{2^{-1/2}} - \frac{1}{\sqrt{2}}^{\rho-1} + (\rho-1)(2^{-1/2})\frac{1}{\sqrt{2}}^{\rho-3}2^{-1/2} \\
&= 2(\rho-1)(2^{-1/2})^{\rho-1}.
\end{aligned}
$$

Observe f is continuous for $-1-\sqrt{2}/2 < \epsilon < 1-\sqrt{2}/2$ Therefore, there exist a $0 < \delta < 1-\sqrt{2}/2$ such that for all $0 < \epsilon < \delta$, $f(\epsilon) > 0$. This proves throughout all of training, $w_1^\top v_1 > w_1^\top v_2$ and similarly for w_2. In words, if the student vectors are initialized closest to different teachers, they will remain closest to their respective teacher throughout all of training.

The only critical point attainable following split teacher initialization is the copy global minimum. Therefore gradient flow will converge to this critical point.

Trivially,

$$
(2^{-1/2}+\epsilon) - \sqrt{\frac{1}{2} - \sqrt{2}\epsilon - \epsilon^2} > 0.
$$

This completes the proof. □